

Tuesday Reading Assessment: Chapter 1

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1 Properties of Digital Pulses

1. A canonical digital pulse is depicted in Fig. 1 (left). Digital components like logic gates have finite capacitance and input impedance that depends on frequency.
 - Suppose the amplitude is CMOS HIGH (between 3.5V and 5V), and has a measured value of 3.55 V. What are the 10% and 90% amplitudes?
 - Suppose we treat the rise as linear, and we label the time of the 10% amplitude as $t = 0 \mu\text{s}$. If the rise occurs at $33 \text{ V}/\mu\text{s}$, when does the 90% amplitude occur? This is the rise time.
 - What is the pulse width, if the pulse width is 5 times the rise time?
 - If the next pulse always occurs two pulse widths from the initial rise, what are the period and frequency of the signal?

2 Logic Operations

1. Three basic logic gates are depicted in Fig. 1 (right).
 - What is the output bit stream from left gate if the input is: 1000 1000?
 - What is the output bit stream from the middle gate if the two input bit streams are: 1000 1000 and 1001 1001?
 - What is the output bit stream from the right gate if the two input bit streams are: 1000 1000 and 1001 1001?

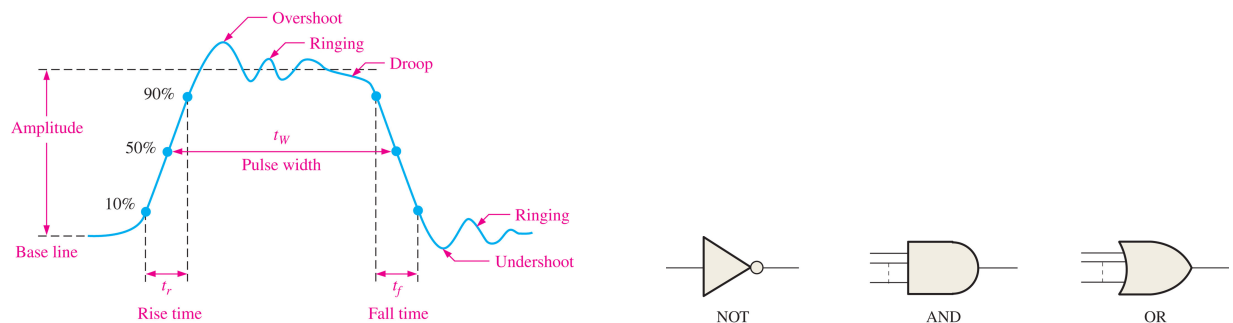


Figure 1: (Left) Properties of a digital pulse. (Right) Three basic logic gates.